

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

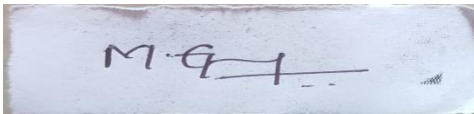
Total Marks: 25

Code of Conduct

I Maddirala Gopi Krishna Reddy 192372055 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Answer:

Handling Seasonal Demand

- Load Balancers act as traffic cops, distributing incoming user requests evenly across multiple backend servers using algorithms like Round-Robin to prevent any single server from overloading.
- Auto-Scaling dynamically adjusts compute capacity by automatically adding

servers (scaling out) during high-traffic spikes and removing them (scaling in) when traffic drops.

Reducing Downtime & Improving User Experience

- Prevents Crashes: Automatically adding capacity before servers reach 100% load prevents HTTP 503 "Service Unavailable" errors.
- High Availability: Load balancers perform continuous health checks; if a server fails, traffic is instantly rerouted to healthy instances with zero downtime.
- Fast Response Times: Keeping server load balanced maintains low latency, ensuring fast page load times and smooth checkouts for buyers.

Role of Monitoring Tools

Monitoring tools (e.g., AWS CloudWatch, Datadog) collect real-time system metrics such as CPU usage and network traffic. When a pre-set threshold is crossed (e.g., CPU load > 75%), the tool triggers an alarm that instructs the auto-scaling group to launch new servers immediately.

Real-World E-Commerce Example

During a Black Friday Flash Sale:

- 11:59 PM: Normal traffic runs on 4 servers.
- 12:00 AM: 200,000 users flood the site at once. Monitoring tools detect a CPU spike and trigger Auto-Scaling to add 16 new instances in minutes.
- Result: The Load Balancer splits traffic across all 20 healthy servers, allowing users to buy items without site crashes. When traffic slows down at 3:00 AM, redundant servers automatically shut down to save costs.

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2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy.

How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

Answer:

Durability and Data Redundancy

- Data Replication: Cloud providers maintain high durability (often 99.999999999% or "11 nines") by replicating data across multiple

physical devices, storage racks, and data centers.

- **Multi-Region Redundancy:** Data can be automatically copied across geographically separate Availability Zones or regions. If an entire data center suffers a natural disaster or power outage, the backup data remains available elsewhere.
- **Erasur Coding & Erasure Self-Healing:** Storage systems break data into encoded fragments spread across multiple disks. The system continuously runs automated integrity checks (bit-rot detection) and automatically heals/rebuilds corrupted data behind the scenes.

Protecting Sensitive Backup Data

- **Encryption (At Rest & In Transit):**
 - **In Transit:** TLS/SSL protocols encrypt backup files while they are transferred from on-premise systems to the cloud, preventing eavesdropping.
 - **At Rest:** Data stored on disks is encrypted using strong algorithms (such as AES-256). Organizations can manage their own encryption keys (KMS) or use provider-managed keys.
- **Access Control & Security:**
 - **Identity and Access Management (IAM):** Enforces the principle of least privilege, allowing only authorized service accounts and administrators to access backup buckets.
 - **Immutability (Object Lock / WORM):** Write-Once-Read-Many policies prevent backup data from being modified or deleted for a set duration, providing crucial protection against ransomware attacks.

Cost Advantages Over Traditional Storage Systems

- **Pay-As-You-Go (OpEx vs. CapEx):** Eliminates massive upfront capital expenses (CapEx) for physical servers, SAN/NAS hardware, and tape libraries in favor of predictable operational expenses (OpEx).
- **Automated Storage Tiering:** Cloud storage offers multiple tiers (e.g., Standard, Frequent Access, Cold/Archive, Glacier Deep Archive). Infrequently accessed backup data can be automatically transitioned to ultra-low-cost archive tiers.
- **No Hardware Maintenance:** Cuts ongoing operational costs related to power, cooling, physical facility footprint, and routine hardware

refreshes or disk replacements.

Real-World Disaster Recovery Use Case

Scenario: On-Premises Ransomware Attack on a Financial Firm

- The Incident: A regional bank suffers a severe ransomware attack that encrypts its primary local databases and renders on-premises servers unusable.
- The Cloud Recovery:
 1. The bank activates its Cloud Disaster Recovery (DR) plan.
 2. Because historical backups were continuously replicated to an immutable cloud storage bucket (AWS S3 Glacier / Azure Blob Storage) with Object Lock enabled, the ransomware was unable to delete or encrypt the cloud backups.
 3. The IT team spins up cloud virtual machines (VMs) in a clean cloud environment and restores the database state directly from the secure cloud backup within hours.
- Outcome: The bank restores normal business operations with minimal downtime and zero data loss, without having to pay a ransom or rebuild local physical infrastructure from scratch.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

Answer:

A Content Delivery Network (CDN) is a geographically distributed network of proxy servers designed to deliver content quickly.

- Caching Dynamic & Static Assets: CDNs store copies of static assets (images, game patches, media files) and cached API responses on servers spread around the world instead of relying solely on a single origin server.
- Distance & Latency Reduction: Latency is heavily influenced by the physical distance data must travel across network hops. By routing requests to the nearest CDN server rather than a distant central data center, Round-Trip Time (RTT) is drastically reduced.
- Offloading Origin Servers: CDNs absorb massive spikes in user traffic,

preventing origin servers from hitting network bandwidth or CPU bottlenecks during global launches.

Role of Edge Servers in Enhancing Regional UX

Edge servers (located at Points of Presence, or PoPs) sit at the "edge" of the network, as close to the end-user as possible.

- **Edge Logic & Compute:** Modern edge networks can execute lightweight code directly on the edge server (e.g., Cloudflare Workers, AWS Lambda@Edge). This allows user authentication, geo-routing, and dynamic content customization to occur locally without contacting the origin database.
- **Network Optimization:** Edge servers maintain persistent, pre-established TCP connections to origin servers, reducing the latency overhead caused by repeated TLS/SSL handshakes over long distances.
- **Consistent Quality Across Regions:** Users in regions far from the application's primary data center receive the same low-latency, fast load times as users located right next to the origin.

How Cloud Providers Support Global Availability

Major cloud providers (e.g., AWS, GCP, Azure) supply the underlying infrastructure to keep mobile apps accessible worldwide:

- **Global Backbone Infrastructure:** Private fiber-optic networks route user traffic securely and directly between edge locations and cloud data centers, bypassing congested public internet routes.
- **Multi-Region Deployments:** Applications can be deployed across multiple Availability Zones (AZs) and regions worldwide to ensure high availability and regional fault isolation.
- **Global Traffic Management (Anycast DNS):** Cloud providers use Anycast IP routing and Latency-Based DNS Routing to automatically direct mobile app users to the nearest healthy application instance.

Real-World Application: Multiplayer Online Game or Live Streaming

Scenario: A company launches a globally distributed multiplayer online mobile game (e.g., PUBG Mobile or Valorant Mobile) or a live video streaming app (e.g., Twitch).

- **Without CDN / Edge Setup:**
 - A player in Tokyo connects to an origin server in Virginia, USA. The high physical distance introduces 200ms+ of network lag (ping).
 - In gaming, this causes game input delays and desync. In live streaming, it leads to severe video buffering and frame drops.
- **With CDN & Cloud Edge Infrastructure:**
 1. **Game Patch Downloads & Video Chunks:** Massive game updates (5 GB

patches) or live HLS/MPEG-DASH video segments are cached directly on Edge Servers local to Tokyo. Users download assets at high-speed local internet rates.

2. Low-Latency Matchmaking & Live Feeds: Player inputs or live video streams are ingested at the nearest Cloud Edge PoP and routed across the cloud provider's low-latency private network directly to the regional game server.
3. Result: Players experience smooth gameplay with sub-30ms ping, and viewers receive high-definition video streams with minimal buffering, regardless of their location.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

Answer:

Regulatory Compliance and Auditing in the Cloud

Cloud providers support financial institutions by operating under a Shared Responsibility Model and offering built-in regulatory frameworks:

- Framework Certifications: Major cloud providers maintain independent compliance certifications for strict standards like PCI DSS (Payment Card Industry Data Security Standard), SOC 1 / SOC 2, ISO 27001, and region-specific frameworks (e.g., DORA in Europe).
- Attestations & Reports: Providers give financial firms access to formal audit evidence—such as Attestations of Compliance (AOC) and third-party SOC 2 Type II audit reports—to present to regulatory bodies.
- Automated Compliance Governance: Cloud platforms offer compliance dashboards (e.g., AWS Security Hub, Azure Policy) that continuously scan cloud resources against regulatory benchmarks and automatically flag policy violations.

Role of Logging, Monitoring, and Encryption

- Logging (Immutable Audit Trails): Centralized logging services (e.g., AWS

CloudTrail, Google Cloud Audit Logs) record every API call, administrative access, and data modification. Write-Once-Read-Many (WORM) storage ensures logs are immutable and tamper-proof for forensic audits.

- **Monitoring (Real-Time Threat Detection):** Security Information and Event Management (SIEM) tools analyze log feeds using machine learning to detect anomalous behavior—such as unauthorized access attempts or suspicious data exfiltration—and trigger real-time security alerts.
- **Encryption (Data Security at All States):**
 - **In Transit:** Protects data moving across networks using TLS 1.3 / SSL.
 - **At Rest:** Encrypts stored databases and backups using AES-256. Financial institutions use Customer-Managed Encryption Keys (CMEK) or Hardware Security Modules (HSM) so that even the cloud provider cannot decrypt the raw data.

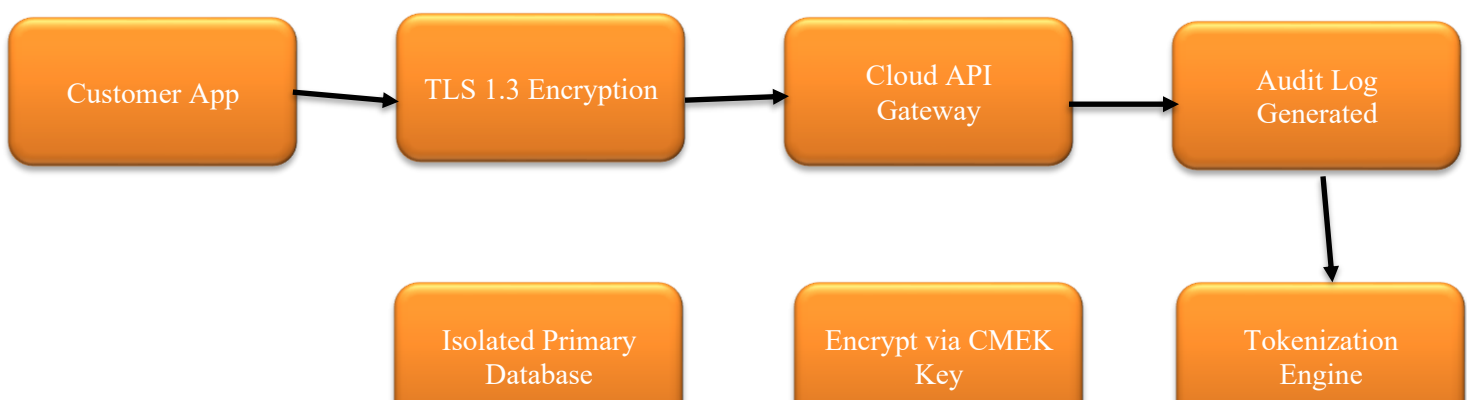
Ensuring Data Sovereignty Requirements

Data Sovereignty mandates that data is subject to the privacy laws and governance of the specific country or region where it is collected or stored. Financial institutions fulfill these requirements through:

- **Geographic Data Residency Pinning:** Selecting specific, fixed physical cloud regions (e.g., AWS eu-central-1 in Frankfurt) and using organization policies to restrict data deployment outside national borders.
- **Sovereign Cloud Solutions:** Deploying dedicated sovereign cloud instances or sovereign controls (e.g., AWS Digital Sovereignty Pledge, Google Cloud Sovereign Cloud) where cloud operators reside in-country and encryption keys are held locally.
- **Client-Side Encryption:** Encrypting financial data on-premises before uploading it to the cloud, ensuring cross-border transfer laws cannot force the cloud host to decrypt sensitive customer records without the bank's key.

Real-World Example: Processing a Credit Card Transaction (PCI DSS Scope)

Scenario: A bank processes online credit card payments via a cloud application.





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5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Answer:

Containers vs. Virtual Machines (Performance & Resource Usage)

While both isolate applications, they operate at different architectural layers:

App A	App B	App C	App A	App B	App C
Guest OS	Guest OS	Guest OS	Container Engine		
Hypervisor			Host Operating System		
Physical Hardwares			Physical Hardwares		

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Virtual Machines (Hardware Virtualization): Each VM includes a full Guest Operating System (OS), virtual kernel, drivers, and application binaries running on top of a Hypervisor.

- Resource Usage: Heavy storage footprint (gigabytes per VM) and high memory overhead due to running multiple Guest OS instances.
- Performance: Slower boot times (minutes) because the entire guest OS must initialize.
- Containers (OS-level Virtualization): Containers share the host system's OS kernel and isolate user space using features like Linux namespaces and cgroups.
 - Resource Usage: Extremely lightweight (megabytes per container) as they only package application code and dependencies.
 - Performance: Near-bare-metal performance with near-instant boot times (seconds or milliseconds).

2. How Orchestration (e.g., Kubernetes) Simplifies Deployment & Scaling

Managing dozens or hundreds of individual containers manually is unsustainable. Container Orchestration automates deployment, management, and networking:

- Automated Scaling (Horizontal Pod Autoscaling): Kubernetes monitors application load (CPU/Memory usage or custom metrics) and automatically adjusts the number of running container instances up or down based on traffic

spikes.

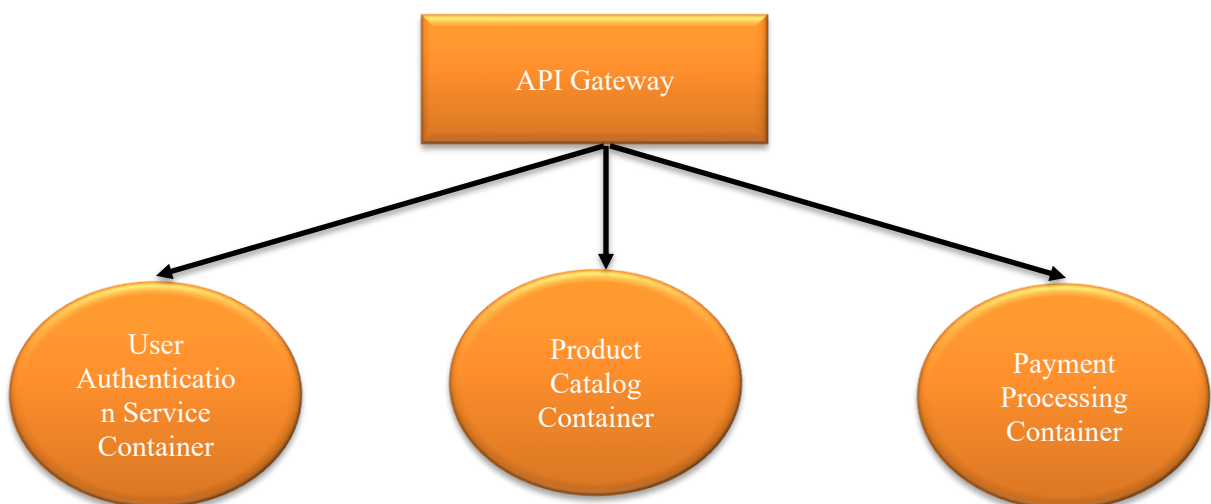
- **Self-Healing & Fault Tolerance:** If a container crashes, fails a health check, or its underlying worker node goes down, Kubernetes automatically terminates and replaces it on a healthy node without human intervention.
- **Load Balancing & Service Discovery:** Kubernetes assigns containers a unique IP address and a single DNS name, distributing network traffic evenly across active container replicas.
- **Automated Rollouts & Rollbacks:** Allows blue-green or canary deployments, replacing old container versions with new ones gradually with zero downtime.

3. Portability Benefits Across Cloud Environments

- **"Build Once, Run Anywhere":** Containers bundle an application with all its specific library versions and runtime dependencies. This eliminates the classic "it works on my machine" problem across development, staging, and production.
- **Multi-Cloud & Hybrid Cloud Deployment:** Because containers run consistently on any host with a container engine (e.g., Docker, containerd), teams can migrate applications seamlessly between AWS, Azure, Google Cloud, or on-premises servers without rewriting code.
- **Vendor Lock-In Mitigation:** Standardizing on open orchestration frameworks like Kubernetes ensures infrastructure can move between cloud providers with minimal friction.

4. Real-World Example: Microservices-Based E-Commerce Application

Instead of building a massive single application (monolith), an online retail platform (e.g., Netflix or Amazon) breaks its system into decoupled microservices, each running in its own container:



Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity